Rigorous Derivation of the Total Derivative

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1 Introduction

This document provides a rigorous proof for the total derivative of a multivariable function $F(x, y(x))$ with respect to x. To ensure a logical flow, we first establish the single-variable Mean Value Theorem and use it to construct the formal definition of multivariable differentiability.

2 Prerequisites

2.1 The Single-Variable Mean Value Theorem

To analyze changes in multiple dimensions, we must first rely on a fundamental property of one-dimensional continuous functions.

Lemma 2.1 (Mean Value Theorem). *If a function $f(t)$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then there exists at least one point $c \in (a, b)$ such that:*

$$f(b) - f(a) = f'(c)(b - a)$$

2.2 The Increment Theorem (Multivariable Differentiability)

In single-variable calculus, differentiability simply requires the limit of the difference quotient to exist. In multivariable calculus, differentiability requires that the function can be closely approximated by a tangent plane, with an error that vanishes rapidly. We derive this required condition using the Mean Value Theorem.

Theorem 2.2 (The Increment of a Function). *Let $F(x, y)$ be a function with continuous partial derivatives $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial y}$ on an open region containing the point (x, y). Let Δx and Δy be small increments. The total change in the function, $\Delta F = F(x+\Delta x, y+\Delta y) - F(x, y)$, can be expressed as:*

$$\Delta F = \frac{\partial F}{\partial x}\Delta x + \frac{\partial F}{\partial y}\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$.

Proof. Consider the total change $\Delta F = F(x+\Delta x, y+\Delta y) - F(x, y)$. We can break this diagonal movement into two orthogonal steps by introducing an intermediate point, $F(x + \Delta x, y)$:

$$\Delta F = [F(x + \Delta x, y + \Delta y) - F(x + \Delta x, y)] + [F(x + \Delta x, y) - F(x, y)]$$

In the first bracket, only the y-variable changes. In the second, only the x-variable changes. We apply the single-variable Mean Value Theorem to each bracket:

1. For the change in x, there exists a point c_1 between x and $x + \Delta x$ such that:

$$F(x + \Delta x, y) - F(x, y) = \frac{\partial F}{\partial x}(c_1, y)\Delta x$$

2. For the change in y, there exists a point c_2 between y and $y + \Delta y$ such that:

$$F(x + \Delta x, y + \Delta y) - F(x + \Delta x, y) = \frac{\partial F}{\partial y}(x + \Delta x, c_2)\Delta y$$

Substituting these back yields:

$$\Delta F = \frac{\partial F}{\partial x}(c_1, y)\Delta x + \frac{\partial F}{\partial y}(x + \Delta x, c_2)\Delta y$$

We want this expression in terms of the derivatives evaluated at our starting point (x, y). We define the error terms ε_1 and ε_2 as the difference between the derivative at the intermediate points and the derivative at the origin point (x, y):

$$\varepsilon_1 = \frac{\partial F}{\partial x}(c_1, y) - \frac{\partial F}{\partial x}(x, y) \implies \frac{\partial F}{\partial x}(c_1, y) = \frac{\partial F}{\partial x}(x, y) + \varepsilon_1$$

$$\varepsilon_2 = \frac{\partial F}{\partial y}(x + \Delta x, c_2) - \frac{\partial F}{\partial y}(x, y) \implies \frac{\partial F}{\partial y}(x + \Delta x, c_2) = \frac{\partial F}{\partial y}(x, y) + \varepsilon_2$$

Substituting these definitions into our equation for ΔF:

$$\Delta F = \left[\frac{\partial F}{\partial x}(x, y) + \varepsilon_1\right]\Delta x + \left[\frac{\partial F}{\partial y}(x, y) + \varepsilon_2\right]\Delta y$$

$$\Delta F = \frac{\partial F}{\partial x}\Delta x + \frac{\partial F}{\partial y}\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y$$

Because we assumed the partial derivatives are continuous, and because $c_1 \to x$ and $c_2 \to y$ as $\Delta x, \Delta y \to 0$, it must follow that $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$. This concludes the proof of the Increment Theorem. $\square$

3 The Total Derivative Proof

With the definition of multivariable differentiability established, we can now prove the chain rule for a path parameterized by x.

Theorem 3.1 (Total Derivative)**.** *Let $F(x, y)$ be continuously differentiable, and let $y = y(x)$ be a differentiable function of x. The total derivative of F with respect to x is given by:*

$$\frac{d}{dx}[F(x, y(x))] = \frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial y}\frac{dy}{dx}$$

Proof. Let x undergo a small change Δx. Because y is a function of x, this induces a corresponding change $\Delta y = y(x + \Delta x) - y(x)$.

From our prerequisites, because F is continuously differentiable, the total change ΔF is exactly:

$$\Delta F = \frac{\partial F}{\partial x}\Delta x + \frac{\partial F}{\partial y}\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y$$

Divide the entire equation by Δx to evaluate the rate of change:

$$\frac{\Delta F}{\Delta x} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\Delta y}{\Delta x} + \varepsilon_1 + \varepsilon_2\frac{\Delta y}{\Delta x}$$

Now, we take the limit as $\Delta x \to 0$:

$$\lim_{\Delta x \to 0} \frac{\Delta F}{\Delta x} = \lim_{\Delta x \to 0} \left[\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\Delta y}{\Delta x} + \varepsilon_1 + \varepsilon_2\frac{\Delta y}{\Delta x}\right]$$

We evaluate the limits of the individual components:

- $\lim_{\Delta x \to 0} \frac{\Delta F}{\Delta x} = \frac{dF}{dx}$ by definition.
- $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$ because $y(x)$ is differentiable.
- Because $y(x)$ is differentiable, it is continuous. Therefore, as $\Delta x \to 0$, $\Delta y \to 0$.
- Because $(\Delta x, \Delta y) \to (0, 0)$, our Increment Theorem dictates that $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$.

Substituting these limits into the equation:

$$\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\left(\frac{dy}{dx}\right) + 0 + 0 \cdot \left(\frac{dy}{dx}\right)$$

$$\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx}$$

Noting that $\frac{dx}{dx} = 1$, we can express this in the requested symmetric form:

$$\frac{dF}{dx} = \frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial y}\frac{dy}{dx}$$

$\square$

4 Self-Check Questions

Q: What specific condition on the partial derivatives of $F(x, y)$ guarantees that the error terms ε_1 and ε_2 vanish in the limit?
A: The partial derivatives must be continuous (i.e., the function is of class C^1). This continuity ensures that the derivative evaluated at an intermediate point (c_1, y) smoothly approaches the derivative at the exact point (x, y) as $\Delta x \to 0$.

Q: During the derivation of the increment ΔF, what is the mathematical purpose of introducing the intermediate point $F(x + \Delta x, y)$?
A: It breaks the simultaneous diagonal shift into two distinct, orthogonal steps (one strictly parallel to the x-axis, one strictly parallel to the y-axis). This isolation is required to apply the single-variable Mean Value Theorem.

Q: When evaluating $\lim_{\Delta x \to 0}$, how do we formally justify that $\Delta y \to 0$ as well?
A: The theorem requires $y = y(x)$ to be a differentiable function. Differentiability strictly implies continuity. By the definition of continuity, a vanishingly small change in the input ($\Delta x \to 0$) forces a vanishingly small change in the output ($\Delta y \to 0$).

Q: State the fundamental difference between the partial derivative $\frac{\partial F}{\partial x}$ and the total derivative $\frac{dF}{dx}$ in the context of this theorem.
A: The partial derivative $\frac{\partial F}{\partial x}$ measures the rate of change of F assuming all other variables (like y) are held strictly constant. The total derivative $\frac{dF}{dx}$ measures the complete rate of change of F as x varies, explicitly accounting for the fact that y is forced to change alongside x because $y = y(x)$.